

ENEL700 Tutorial 5

1. Determine the Nyquist Rate and Nyquist Sampling Interval for the following signals: (1) $5 \cos(200\pi t)$ (2) $20 - \sin^2(10^4 \pi t)$ (3) $10 \text{rect}(2 \times 10^3 t)$

(i) $f_m = 200\pi / 2\pi = 100 \text{ Hz}$

→ Nyquist Rate = $2f_m = \underline{200 \text{ Hz}}$; Sampling Interval = $1/200 = \underline{5 \text{ ms}}$

(ii) Using trigonometric identity: $\cos 2\theta = 1 - 2\sin^2 \theta$

$$20 - \sin^2(10^4 \pi t) = 20 - \frac{1 - \cos(2 \times 10^4 \pi t)}{2} = 19.5 + \frac{\cos(2 \times 10^4 \pi t)}{2}$$

$f_m = 2 \times 10^4 \pi / 2\pi = 10 \text{ kHz}$

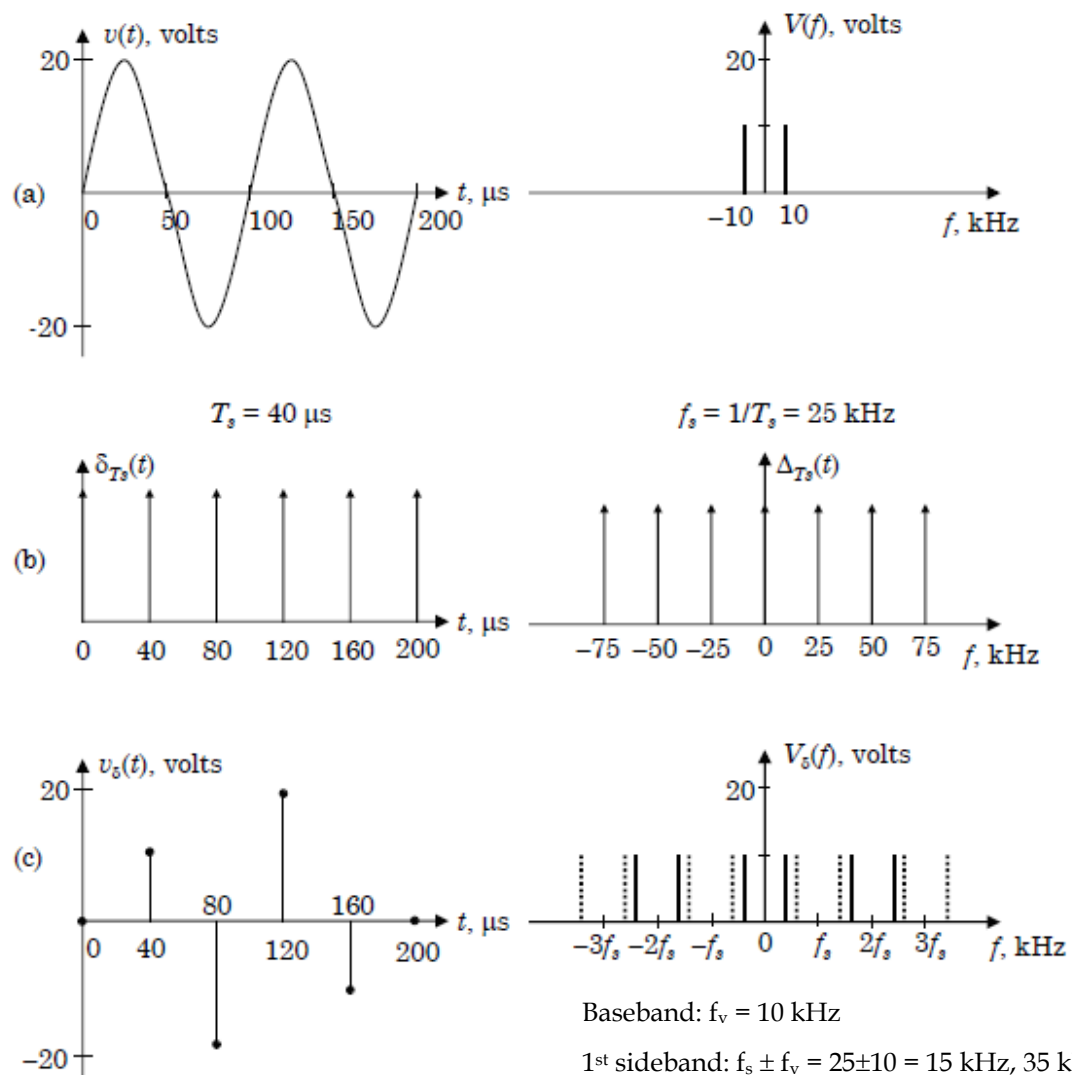
→ Nyquist Rate = $\underline{20 \text{ kHz}}$; Sampling Interval = $1/20000 = \underline{50 \text{ us}}$

- (iii) The signal is a rectangular pulse expressed by equation $10 \text{rect}(t/\tau)$. Thus, it has a height 10 and duration $\tau = 0.5 \text{ ms}$.

Maximum (significant) frequency component is given by: $f_m = 1/\tau = 2 \text{ kHz}$

→ Nyquist Rate = $4 \underline{\text{kHz}}$; Sampling Interval = $1/4000 = 0.25 \underline{\text{ms}}$

2. A sinusoidal voltage signal $v(t) = 20\sin(2\pi \times 10^4 t)$ volts is sampled using an impulse train $\delta_{T_s}(t)$ of period $T_s = 40\mu\text{s}$. Sketch the waveform of corresponding double-sided amplitude spectrum of the following signals: (1) $v(t)$ (2) $\delta_{T_s}(t)$ (3) Sampled signal $v_s(t)$ (should contain at least three sidebands).



Baseband: $f_v = 10\text{ kHz}$

1st sideband: $f_s \pm f_v = 25 \pm 10 = 15\text{ kHz}, 35\text{ kHz}$

2nd sideband: $2f_s \pm f_v = 2(25) \pm 10 = 40\text{ kHz}, 60\text{ kHz}$

3rd sideband: $3f_s \pm f_v = 3(25) \pm 10 = 65\text{ kHz}, 85\text{ kHz}$

n^{th} sideband: $nf_s \pm f_v$, where $n = 1, 2, 3, \dots$

3. A signal that ranges from 0 to +3V must be converted to digital form with a quantisation error of less than 1mV. How many bits per sample would an ADC have to provide to meet this goal?

$$\text{Maximum quantisation error} = \Delta/2 \leq 1 \text{ mV} \rightarrow \Delta \leq 2 \text{ mV}$$

$$\Delta = \frac{2C}{2^K} = \frac{V_{full-range}}{2^K} = \frac{3}{2^K} \leq 2 \text{ mV}$$

$$\rightarrow 2^K \geq \frac{3}{2 \times 10^{-3}} = 1500$$

$$\rightarrow K \geq \frac{\log_{10} 1500}{\log_{10} 2} = 10.55$$

Hence, ADC has to provide $K = 11$ bits per sample.

4. The output of a midrise uniform quantiser of input range $\pm 5V$ is coded using 8 bits/sample. Determine: (1) the maximum quantisation error; (2) the quantisation error associated with an input sample of value $-2.3V$; (3) the quantisation noise power; (4) the dynamic range of the system; (5) the SQNR when the input is a speech signal ($R=2.818$) that fully loads the quantiser; (6) the SQNR during a weak passage when the peak value of the input speech signal is only $0.5V$.

(i) Step size $\Delta = \frac{2C}{N} = \frac{2 \times 5}{2^8} = 39.0625 \text{ mV}$

→ Maximum quantisation error $= \frac{\Delta}{2} = 19.53 \text{ mV}$

- (ii) The input sample $s = -2.3 \text{ V}$ falls in the l^{th} interval (counting from 0 to $N-1$ between $-C$ and $+C$), where $l = \left\lfloor \frac{s - (-C)}{\Delta} \right\rfloor = 69$

The quantised output s_q is the midpoint of the l^{th} interval:

$$s_q = -C + l\Delta + \frac{\Delta}{2} = -2.285 \text{ V}$$

The quantisation error for this sample is therefore:

$$q = |s - s_q| = 15 \text{ mV}$$

- (iii) The MSQE gives the quantisation noise power: $MSQE = \frac{\Delta^2}{12} = 127.2 \text{ } \mu W$

- (iv) Dynamic range $= 6.02K = 48.16 \text{ dB}$

- (v) $SQNR = 4.77 + 6.02K - 20\log_{10}(R) + 20\log_{10}(V_p/C)$
 With $V_p = C \rightarrow SQNR = 6.02K - 4.23 = 43.93 \text{ dB}$

- (vi) $SQNR = 6.02K - 4.23 + 20\log_{10}(V_p/C)$
 With $V_p = 0.5, C = 5 \rightarrow SQNR = 23.93 \text{ dB}$